

Operation and Maintenance Manual

Diesel engine-generator sets

DP00365D5S

DP00400D5S

DS00400D5S

DS00440D5S

Application groups 3B, 3D

Built in Europe

Control versions 2, 3a, 3b, 4

SS180013/02E



System designation	Engine model	Application group
DP00365D5S	8V1600G10F	3B, Continuous operation, variable
DP00400D5S	8V1600G20F	3B, Continuous operation, variable
DS00400D5S	8V1600G70F	3D, Short-term operation, fuel stop
DS00440D5S	8V1600G80F	3D, Short-term operation, fuel stop

Table 1: Applicability

Preface

This Operation & Maintenance Manual provides general instructions for operating your MTU Onsite Energy engine-generator set properly. It is essential that every person who works on or with the engine-generator set be completely familiar with the contents of this manual and that he/she carefully follows the instructions contained herein.

Each operation and maintenance may require some modification of the suggested guidelines in this manual. They must be consistent with locally applicable standards and take into consideration safety guidelines and measures.

Following this guide will result in an efficient and reliable installation.

Carefully follow all procedures and safety precautions to ensure proper equipment operation and to avoid bodily injury. Read and follow the Safety section at the beginning of this manual.

All instructions and diagrams have been checked for accuracy and simplicity of application, however, the skills of the operator are most important. MTU Onsite Energy does not guarantee the result of any operation contained in this manual nor can MTU Onsite Energy assume responsibility for any injury or damage to property. Persons engaging in operating do so entirely at their own risk.

In this publication the manufacturer of the engine-generator set is known as MTU Onsite Energy.

Table of Contents

1 Safety			
1.1 Important requirements for all products	8	2.6.9 Day tank	70
1.2 Personnel and organizational requirements	9	2.6.10 Sub-base fuel tank	72
1.3 Safety requirements for startup and operation	10	2.7 Air System	74
1.4 Safety requirements for maintenance and repair work	11	2.7.1 Air system	74
1.5 Fire prevention and environmental protection, fluids and lubricants, auxiliary materials	14	2.7.2 Air filter	76
1.6 Standards for safety notices in the text	16	2.7.3 Heavy duty air filter - Mounting location	78
		2.7.4 Heavy duty air filter	79
2 Functional Description		2.8 Exhaust System	81
2.1 Chapter overview	17	2.8.1 Exhaust system	81
2.2 Engine-Generator Set	18	2.8.2 Exhaust silencer	83
2.2.1 Engine-generator set - General information	18	2.9 Cooling System	85
2.2.2 Engine-generator set - Scope of supply	20	2.9.1 Cooling system	85
2.3 Engine	23	2.9.2 Coolant cooler	87
2.3.1 MTU Series 1600 GxO engine	23	2.9.3 Preheater - Mounting location	89
2.3.2 Sensors and actuators	25	2.9.4 Preheater	90
2.3.3 Engine side and cylinder designations	27	2.10 Electrical System	92
2.3.4 Firing order	28	2.10.1 Starter battery	92
2.4 Generator	29	2.10.2 Battery charger	94
2.4.1 Generator - General information	29	2.10.3 Circuit breaker	96
2.4.2 Digital voltage regulator DVR2000E+	32	2.11 Base and mounting system	99
2.5 Monitoring and Control	36	2.12 Enclosure	100
2.5.1 Control version 2 - Isolated operation (without mains)	36		
2.5.2 Control version 3a - With automatic transfer switch	38	3 Technical Data	
2.5.3 Control version 3b - With electrical interlock circuit breakers	40	3.1 Technical data DP00365D5S (fuel optimized)	103
2.5.4 Control version 4 - Parallel operation (island)	42	3.2 Technical data DP00365D5S (exhaust optimized)	105
2.5.5 Control cabinet with DGC-2020 - General information	44	3.3 Technical data DP00400D5S (fuel optimized)	107
2.5.6 Control panel DGC-2020 - Display operations	50	3.4 Technical data DP00400D5S (exhaust optimized)	109
2.5.7 Smart Connect	54	3.5 Technical data DS00400D5S (fuel optimized)	111
2.5.8 Read out the active control version on DGC-2020	56	3.6 Technical data DS00440D5S (fuel optimized)	113
2.6 Fuel System	57		
2.6.1 Fuel system	57	4 Transport	
2.6.2 Fuel prefilter - Mounting location	59	4.1 Transport	115
2.6.3 Fuel prefilter	60		
2.6.4 Fuel prefilter with water separator - Mounting location	62	5 Operation	
2.6.5 Fuel prefilter with water separator (single)	64	5.1 Chapter overview	117
2.6.6 Fuel prefilter with water separator (double)	66	5.2 Lockout/tagout procedure	118
2.6.7 Fuel cooler - Mounting location	68	5.3 Unlocking procedure	121
2.6.8 Fuel cooler	69	5.4 Putting the engine-generator set into operation	123
		5.5 Start procedures - Overview	126
		5.6 Stop procedures - Overview	127

5.7 Start and Stop Procedures for Control Version 2	128	8.3.3 Cylinder head cover – Removal and installation	268
5.7.1 Automatic start and stop	128	8.3.4 Fuel system – Venting	269
5.7.2 Manual start	130	8.3.5 Fuel filter – Replacement	270
5.7.3 Manual stop	131	8.3.6 Engine oil – Level check	271
5.8 Start and Stop Procedures for Control Version 3a	132	8.3.7 Engine oil – Change	272
5.8.1 Automatic start and stop with ATS	132	8.3.8 Engine oil filter – Replacement	274
5.8.2 Manual start	134	8.3.9 Engine coolant – Level check	275
5.8.3 Manual stop	135	8.3.10 Engine coolant – Change	276
5.9 Start and Stop Procedures for Control Version 3b	136	8.3.11 Engine coolant – Draining	277
5.9.1 Automatic start and stop with electrical interlock circuit breakers	136	8.3.12 Engine coolant – Filling	279
5.9.2 Manual start	139	8.3.13 Engine coolant pump – Relief bore check	281
5.9.3 Manual stop	140	8.3.14 Drive belt – Adjustment	282
5.10 Start and Stop Procedures for Control Version 4	141	8.3.15 Drive belt – Condition check	283
5.10.1 Automatic start and stop	141	8.3.16 Drive belt – Tension check	284
5.10.2 Manual start	143	8.3.17 Drive belt – Replacement	286
5.10.3 Manual stop	145	8.3.18 Engine wiring – Check	288
5.11 Emergency stop	146	8.3.19 Engine governor and connectors – Cleaning	289
5.12 Exercising checks	147	8.3.20 Engine governor – Checking plug connections	290
5.13 Putting the engine-generator set out of operation	149	8.4 Engine-Generator Set Control Panel	291
6 Troubleshooting		8.4.1 Control panel DGC-2020 – Backup battery replacement	291
6.1 Chapter overview	150	8.5 Generator	292
6.2 Troubleshooting by symptoms	151	8.5.1 Generator – Check	292
6.3 Engine-Generator Set Fault Codes	156	8.5.2 Generator – Wiring check	293
6.3.1 Recording of fault codes on DGC-2020	156	8.5.3 Generator – Greasing antifriction bearing	294
6.3.2 List of fault codes on DGC-2020 in alphabetical order	158	8.5.4 Generator mounting securing screws – Security check	296
6.4 Engine Fault Codes	216	8.6 Fuel System	297
6.4.1 Fault codes on Smart Connect display	216	8.6.1 Fuel prefilter with water separator (double) – Differential pressure check	297
6.4.2 List of MTU fault codes in numerical order (ADEC-ECU8)	217	8.6.2 Fuel prefilter – Draining overview	298
7 Maintenance		8.6.3 Fuel prefilter – Draining	299
7.1 Maintenance task reference table [QL1]	261	8.6.4 Fuel prefilter with water separator – Draining	300
8 Task Description		8.6.5 Fuel prefilter – Replacement overview	301
8.1 Chapter overview	262	8.6.6 Fuel prefilter – Replacement	302
8.2 Engine-Generator Set	263	8.6.7 Fuel prefilter with water separator – Filter element replacement	303
8.2.1 Engine-generator set – Test run	263	8.7 Air System	305
8.3 Engine	264	8.7.1 Service indicator – Signal ring position check	305
8.3.1 Engine – Barring with starting system	264	8.7.2 Air filter – Replacement	306
8.3.2 Valve clearance – Check and adjustment	265	8.7.3 Air filter – Removal and installation	307
		8.7.4 Heavy duty air filter – Removal and installation	308
		8.8 Cooling System	309
		8.8.1 Coolant cooler – Check for leaks and contamination	309
		8.8.2 Coolant cooler – Core face cleaning	310
		8.8.3 Preheater – Coolant line check	312
		8.8.4 Preheater – Heating element housing wiring check	313
		8.9 Electrical System	314
		8.9.1 Circuit breaker – Check and function test	314

9 Appendix A		9.4 MTU contact persons/service partners	339
9.1 Abbreviations	316	10 Appendix B	
9.2 Conversion tables	319	10.1 Special Tools	340
9.3 Glossary	323	10.2 Consumables	343
		10.3 Index	344

1 Safety

1.1 Important requirements for all products

Nameplate

The product is identified by a nameplate, model designation or serial number which must match the information on the title page of this manual.

Nameplate, model designation or serial number can be found on the product.

General information

This product may pose a risk of injury or damage in the following cases:

- Improper use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications
- Noncompliance with the safety instructions and warning notices

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (including engine management/parameters)
- In compliance with all safety regulations and in accordance with all warning notices in this manual
- With maintenance work performed in accordance with the (→ Maintenance Schedule) throughout the useful life of the product
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair
- By contracting only workshops authorized by the manufacturer to carry out repair and overhaul

Any other use shall be considered non-intended. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

Changes or modifications

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

TM-ID: 0000040530 - 004

1.2 Personnel and organizational requirements

Organizational measures of the operator

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use this manual as a basis for instructing personnel on product operation and repair, whereby the safety-relevant instructions, in particular, must be read and understood.

This is particularly important in the case of personnel who only occasionally perform work on or around the product. This personnel must be instructed repeatedly.

Personnel requirements

All work on the product shall be carried out by trained and qualified personnel only.

- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

Working clothes and personal protective equipment

Wear proper protective clothing for all work.

When working, always wear the necessary personal protective equipment (e.g. ear protectors, protective gloves, goggles, breathing protection). Observe the information on personal protective equipment in the respective activity description.

1.3 Safety requirements for startup and operation

Safety requirements for startup

The product must be installed and accepted in accordance with manufacturers' specifications prior to initial startup.

All the requisite regulatory approvals must have been granted and all startup requirements fulfilled prior to initial startup.

Whenever the product is started, always ensure that:

- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- Wearers of cardiac pacemakers or other active medical implants are well clear.
- The operating room is adequately ventilated.
- Exhaust pipework is leak-tight and routed to atmosphere.
- Battery terminals, generator terminals and cables are guarded to preclude accidental contact.
- All personnel is clear of the danger zone represented by moving parts.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems are working properly.

Safety requirements for operation

The operator must be familiar with the controls and displays.

The operator must be aware of the consequences of any operations he/she performs.

During operation, the display instruments and monitoring units must be constantly observed in regard of present operating status, limit value violation and warning or alarm messages.

Malfunctions and emergency stop

Emergency procedures, in particular emergency stop, must be practiced on a regular basis.

Take the following steps if a system malfunction is detected or signaled by the system:

- Inform the duty supervisor(s).
- Evaluate the message.
- Respond to the emergency appropriately, e.g. execute an emergency stop.

Operation

Do not stay in the operating room when the product is running unless absolutely necessary and then as briefly as possible.

Keep a safe distance away from the product whenever possible. Do not touch the product unless expressly instructed to do so.

Do not inhale exhaust gases emitted by the product.

Ensure that the following requirements have been fulfilled before starting the product:

- Ear protectors are worn.
- Mop up any leaked or spilled fluids and lubricants immediately or soak up with a suitable binding agent.

Operation of electrical equipment

Some components are live (high voltage) when electrical equipment is in operation.

Observe the safety instructions for these appliances.